

# Tutorial: Aggregate Expenditure Model - Numerical Exercises with Solutions

Macroeconomics: ECON 101

March 4, 2025

## Introduction

The Aggregate Expenditure (AE) Model is a fundamental tool in macroeconomics for analyzing equilibrium output. It is given by:

$$AE = C + I + G + NX$$

where:

- $C$  = Consumption
- $I$  = Investment
- $G$  = Government Spending
- $NX$  = Net Exports (Exports - Imports)

The equilibrium GDP ( $Y^*$ ) is found where  $AE = Y$ . The following exercises include calculations related to equilibrium output, the multiplier effect, fiscal policy, and economic shocks.

## Exercise 1: Finding Equilibrium GDP

**Given:**

$$C = 200 + 0.8Y$$

$$I = 150$$

$$G = 250$$

$$NX = 50$$

**Solution:** The equilibrium condition is:

$$Y = C + I + G + NX$$

Substituting the given values:

$$Y = (200 + 0.8Y) + 150 + 250 + 50$$

$$Y - 0.8Y = 650$$

$$0.2Y = 650$$

$$Y^* = \frac{650}{0.2} = 3250$$

If investment increases to  $I = 200$ , we recalculate:

$$Y = (200 + 0.8Y) + 200 + 250 + 50$$

$$0.2Y = 700$$

$$Y^* = \frac{700}{0.2} = 3500$$

## Exercise 2: The Multiplier Effect

**Given:**

- $MPC = 0.75$
- Autonomous Spending = 500

**Solution:** The spending multiplier is:

$$Multiplier = \frac{1}{1 - MPC} = \frac{1}{1 - 0.75} = \frac{1}{0.25} = 4$$

If government spending increases by 100:

$$\Delta Y = Multiplier \times \Delta G = 4 \times 100 = 400$$

Thus, GDP increases by 400.

### Exercise 3: Fiscal Policy and Aggregate Expenditure

Given:

$$\begin{aligned}C &= 300 + 0.6Y \\I + G + NX &= 900\end{aligned}$$

**Solution:** Equilibrium condition:

$$Y = (300 + 0.6Y) + 900$$

$$Y - 0.6Y = 1200$$

$$0.4Y = 1200$$

$$Y^* = 3000$$

If  $G$  increases by 200:

$$Y = (300 + 0.6Y) + 1100$$

$$0.4Y = 1400$$

$$Y^* = 3500$$

Total change in GDP:

$$\Delta Y = 3500 - 3000 = 500$$

## Exercise 4: Taxes and the Aggregate Expenditure Model

Given:

$$\begin{aligned}C &= 100 + 0.75(Y - 200) \\I &= 150, \quad G = 250, \quad NX = 50\end{aligned}$$

**Solution:** Equilibrium:

$$Y = (100 + 0.75(Y - 200)) + 150 + 250 + 50$$

$$Y - 0.75Y = 500 - 150$$

$$0.25Y = 350$$

$$Y^* = 1400$$

If  $T = 250$ , solve for new  $Y$ :

$$Y = (100 + 0.75(Y - 250)) + 150 + 250 + 50$$

$$0.25Y = 325$$

$$Y^* = 1300$$

## Exercise 5: Savings and the Paradox of Thrift

Given:

$$C = 200 + 0.8Y$$

$$I = 100, \quad G = 300, \quad NX = 0$$

**Solution:** Equilibrium condition:

$$Y = (200 + 0.8Y) + 100 + 300$$

$$0.2Y = 600$$

$$Y^* = 3000$$

If  $MPC = 0.7$ :

$$Y = (200 + 0.7Y) + 100 + 300$$

$$0.3Y = 600$$

$$Y^* = 2000$$

**Interpretation:** A decrease in MPC reduces equilibrium output, illustrating the **Paradox of Thrift**.

## Exercise 6: The Effect of an Export Boom

Given:

$$C = 250 + 0.9Y$$

$$I = 300, \quad G = 450, \quad NX = 50$$

Solution:

$$0.1Y = 1000$$

$$Y^* = 10000$$

If  $NX = 150$ :

$$0.1Y = 1100$$

$$Y^* = 11000$$

## Exercise 7: Balanced Budget Multiplier

Given:

- $MPC = 0.8$
- $\Delta G = \Delta T = 100$

**Solution:** Balanced budget multiplier is:

$$\frac{1}{1 - MPC} - \frac{MPC}{1 - MPC} = 1$$

Thus, GDP increases by exactly 100.



## Exercise 8: A Negative Demand Shock

Given:

$$C = 300 + 0.75Y$$

$$I = 250, \quad G = 400, \quad NX = 50$$

Solution:

$$0.25Y = 1000$$

$$Y^* = 4000$$

If  $I$  drops by 100:

$$0.25Y = 900$$

$$Y^* = 3600$$